Random Variable: Suppose S is the sample space of a given random experiment. A real valued function define on the sample space and taking value from $-\infty$ to $+\infty$ is known as random variable. Random variable is denoted by capital letter X,Y,Z,… etc . Spectrum of a random variable means all possible values of the random variable. Random variable is of two types.

i) **Discrete Random Variable:** A random variable which takes countable finite number of values or which is defined on a discrete sample space.

Example: A coin is tossed three times. Let X denotes number of times head is occurring. Sample space consists of 8 sample points as

Sample point:	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
X :	3	2	2	1	2	1	1	0

Random Variable X will take value 0, 1, 2 or 3.

X : 0 1 2 3

ii) **Continuous Random Variable:** A random variable which takes all possible values in a given range is known as continuous random variable.

Probability mass function: Suppose X is a discrete random variable taking countably infinite number of values $x_1, x_2, x_3, \ldots$ With each possible outcome x_i, we associate a number $p_i = P(X = x_i)$, such that it must satisfy the following conditions

i) $Pi \geq 0$

ii) $\sum p_i = 1$

Then p_i is known as probability mass function of random variable X

- **Distribution function:** It indicates the probability of a random variable from $-\infty$ to a value of that random variable.It is denoted by $F(x) = P(X \leq x)$

Suppose a coin is tossed three times. Let X denotes number of times head is occurring. Then probability distribution of X is

X :	**0**	**1**	**2**	**3**
P(X = x):	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$
F(x) :	$\frac{1}{8}$	$\frac{4}{8}$	$\frac{7}{8}$	1

Example : Suppose X is a random variable taking values

X	-2	-1	0	1	2
P(X =x)	K	2k	3k	3k	k

i) Find value of k such that it is a probability mass function

ii) $P(-1 < X \leq 1)$

iii) $P(-1 \leq X \leq 1)$

iv) $P(-1 < X < 1)$

Solution:

i) If X is a random variable, the sum of all probabilitities must be equal to 1 i.e.

$k + 2k + 3k + 3k + k = 1$

$10\,k = 1$

$k = \frac{1}{10}$

X	-2	-1	0	1	2
P(X =x)	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{3}{10}$	$\frac{1}{10}$

i) $P(-1 < X \leq 1) = P(X = 0) + P(X = 1)$

$$= \frac{3}{10} + \frac{3}{10} = \frac{6}{10}$$

ii) $P(-1 \leq X \leq 1) = P(X = -1) + P(X = 0) + P(X = 1)$

$$= \frac{2}{10} + \frac{3}{10} + \frac{3}{10} = \frac{8}{10}$$

iii) $P(-1 < X < 1) = P(X = 0) = \frac{3}{10}$

Example : Random variable X has following probability distribution

X	-3	-2	-1	0	1	2	3
P(X = x)	K	2k	2k	3k	k^2	$2\,k^2$	$7k^2+k$

i) Find value of k

ii) $P(X < 0)$

iii) $P(-2 < X \leq 1)$

iv) $P(-1 \leq X \leq 2)$

v) $P(-1 < X < 2)$

Solution: i) If X is a random variable, the sum of all probabilities must be equal to 1

i.e. $k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$

$10k^2 + 9k = 1$

$10k^2 + 9k - 1 = 0$

$10k^2 + 10k - k - 1 = 0$

$10k(k + 1) - 1(k + 1) = 0$

$(k + 1)(10k - 1) = 0$

$k + 1 = 0 \Rightarrow k = -1$ or $10k - 1 = 0 \Rightarrow k = \frac{1}{10}$

$k = -1$ rejected as probability cannot be negative, hence $k = \frac{1}{10}$

X	-3	-2	-1	0	1	2	3
P(X = x)	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{1}{100}$	$\frac{2}{100}$	$\frac{17}{100}$

ii) $P(X<0) = P(X=-3) + P(X=-2) + P(X=-1) = \frac{1}{10} + \frac{2}{10} + \frac{2}{10} = \frac{5}{10}$

iii) $P(-2 < X \le 1) = P(X=-1) + P(X=0) + P(X=1) = \frac{2}{10} + \frac{3}{10} + \frac{1}{100} = \frac{51}{100}$

iv) $P(-1 \le X \le 2) = P(X=-1) + P(X=0) + P(X=1) + P(X=2)$

$= \frac{2}{10} + \frac{3}{10} + \frac{1}{100} + \frac{2}{100} = \frac{53}{100}$

v) $P(-1 < X < 2) = P(X=0) + P(X=1) = \frac{3}{10} + \frac{1}{100} = \frac{31}{100}$

Example 4.31: A box contains 10 fruits of which 3 are spoiled. A sample of 4 fruits is drawn from box at random without replacement. Let X denotes number of spoiled fruits. Find probability distribution of X.

Solution: Total Fruits = 10

Spoiled fruit =3

Good fruits $= 10 - 3 = 7$

Now 4 fruits will be selected from the box, spoiled fruits may be 0,1,2,3

$$P(X=0) = \frac{{}^{7}C_4}{{}^{10}C_4} = \frac{\frac{7!}{4!\times 3!}}{\frac{10!}{4!\times 6!}} = \frac{\frac{7\times 6\times 5\times 4!}{3\times 2\times 1\times 4!}}{\frac{10\times 9\times 8\times 7\times 6!}{4\times 3\times 2\times 1\times 6!}} = \frac{35}{210} = \frac{1}{6}$$

$$P(X=1) = \frac{{}^{7}C_3 \times {}^{3}C_1}{{}^{10}C_4} = \frac{\frac{7!}{4!\times 3!}\times 3}{\frac{10!}{4!\times 6!}} = \frac{\frac{7\times 6\times 5\times 4!}{3\times 2\times 1\times 4!}\times 3}{\frac{10\times 9\times 8\times 7\times 6!}{4\times 3\times 2\times 1\times 6!}} = \frac{35\times 3}{210} = \frac{1}{2}$$

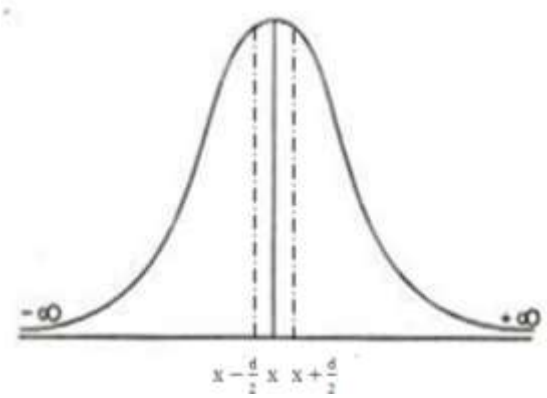
$$P(X=2) = \frac{{}^{7}C_2 \times {}^{3}C_2}{{}^{10}C_4} = \frac{21\times 3}{210} = \frac{3}{10}$$

$P(X=3) = \frac{^7c_1 \times {}^3c_3}{^{10}c_4} = \frac{7\times1}{210} = \frac{1}{30}$

X	0	1	1	3
P(X = x)	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$
F (X)	$\frac{1}{6}$	$\frac{2}{3}$	$\frac{29}{30}$	1

Probability density function (pdf): Let X be a continuous random variable. Take small interval of length dx around the point x and f(x) be any continuous function of x such that $P(x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2}) = f(x)dx$, f(x) is known as probability density function of random variable such that

i) $f(x) > 0$ ii) $\int_{-\infty}^{\infty} f(x)dx = 1$ $-\infty < X < \infty$



Example : Diameter of fruit is given by

$6x(1-x)$ $0 < X < 1$

i) Check that it is a probability density function
ii) Find probability that diameter of fruit is < 0.5
iii) Find probability that diameter of fruit lies between 0.25 to 0.5

Solution: i) $\int_0^1 6x(1-x)\,dx = 1$

$= \int_0^1 6x\,dx - \int_0^1 6x^2\,dx$

$= 6\left[\frac{x^2}{2}\right]_0^1 - 6\left[\frac{x^3}{3}\right]_0^1$

$= 3(1^2 - 0^2) - 2(1^3 - 0^3)$

$= 3 - 2$

= 1

Hence it is a pdf.

ii) P(X < 0.5)

$\int_0^{0.5} 6x(1-x)\,dx$

$\int_0^{0.5} 6x\,dx - \int_0^{0.5} 6x^2\,dx$

$6\left[\frac{x^2}{2}\right]_0^{0.5} - 6\left[\frac{x^3}{3}\right]_0^{0.5}$

$3(0.5^2 - 0^2) - 2(0.5^3 - 0^3)$

$3 \times 0.25 - 2 \times 0.125$

$0.75 - 0.25 = 0.5$

iii) P(0.25 < X < 0.5)

$3(0.5^2 - 0.25^2) - 2(0.5^3 - 0.25^3)$

$3(0.25 - 0.0625) - 2(0.125 - 0.015625)$

$3 \times 0.1875 - 2 \times 0.10937$

$0.5625 - 0.21875$

0.34375

Probability Distribution

Binomial Distribution: Suppose n independent trials are performed repeatedly and each trial results into two mutually exclusive events, occurrence of an event is known as success and non-occurrence of an event is known as failure. Probability of success remains constant from trial to trial. If there are x successes, then there will be n-x failure, these success and failure may occured in any order

SFFFSSFFSSFSFSS…

- Let probability of success is p and probability of failure is q, then probability of x successes in n independent trial is

  pqqqppqqppqpqpp…

  $= p^x.q^{n-x}$

  But x successes in n trials occurs in ${}^nc_x$ ways

  Hence probability of x successes in n independent trial is given by ${}^nc_x.\ p^x.q^{n-x}$

- A random variable X is having binomial distribution with parameters n and p and its probability mass function (p.m.f.) is given by

  $P(X = x) = {}^nc_x.\ p^x.q^{n-x}$

  $x = 0,1, 2, 3,\ldots, n$

  $0 \le p \le 1$

  $p + q = 1$

  Also $\sum {}^nc_x.\ p^x.q^{n-x} = (q + p)^n = 1$

Physical conditions of binomial distribution:

i) Number of trials (n) are finite

ii) Each trial is independent to each other

iii) Each trial results in two mutually exclusive events i.e. success or failure

iv) Probability of success remains constant from trial to trial

v) It is generally applied when sampling is done with replacement from infinite population so that probability of success remains same.

A box contains 100 fruits of which 30 fruits are damaged. Six fruits are selected randomly, what is the probability that it contains

i) Exactly 2 damaged fruits

ii) No damaged fruit

iii) Atleast 1 damaged fruit

iv) Atmost 5 damaged fruits

Solution: Probability of damaged fruit = p = $\frac{30}{100}$ = 0.3, q = 1- 0.3 = 0.7

i) Exactly 2 fruits are damaged

$P(X = 2) = {}^6C_2 \times 0.3^2 \times 0.7^{(6-2)}$

$= 15 \times 0.09 \times 0.2401 = 0.324$

ii) No damaged fruit

$P(X = 0) = {}^6C_0 \times 0.3^0 \times 0.7^6$

$= 0.7^6 = 0.117$

iii) Atleast one damaged fruit

$= 1 - P(X = 0) = 1 - 0.117649 = 0.882$

iv) Atmost five damaged fruits

$= 1 - P(X = 6)$

$= 1 - {}^6C_6 \times 0.3^6 \times 0.7^{6-6}$

$= 1 - 0.3^6 = 1 - 0.000729 = 0.9993$

Constants of binomial distribution:

Mean	=	np
Variance (μ_2)	=	npq

Note : Mean of binomial distribution is always greater than variance

- **Mode :** There are two cases
  - i) If (n+1).p is not an integer, distribution is unimodal and integer part of (n+1).p will be the value of mode
  - ii) If (n+1).p is an integer, distribution is bi-modal, let (n+1).p = m, two mode will be m − 1 and m.

Example : Mean of binomial distribution is 4 and variance is 3. Find $P(X \geq 2)$ and mode of the distribution

Solution: Mean = np = 4 (i)

Variance = npq = 3 (ii)

Divide (ii) by (i)

$\frac{npq}{np} = \frac{3}{4}$

$q = \frac{3}{4}$, $p = 1 - \frac{3}{4} = \frac{1}{4}$

$n.\frac{1}{4} = 4$

$n = 16$

Hence parameters of binomial distribution are B ($n = 16$, $p = \frac{1}{4}$)

$P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$

$P(X = 0) = {}^{16}C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{16} = \left(\frac{3}{4}\right)^{16} = 0.010023$

$P(X = 1) = {}^{16}C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{16-1} = 16 \times \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{16-1} = 0.053454$

$P(X \geq 2) = 1 - 0.010023 - 0.053454$

$= 0.936524$

Mode: $(n+1).p = (16+1) \times \frac{1}{4} = \frac{17}{4} = 4.25$

Since 4.25 is in fraction and its integer part is 4, hence mode of distribution is 4.

Example : Mean of binomial distribution is 3 and variance is 4. Find parameter of the distribution

Solution: Mean = np = 3 ..... (i)

Variance = npq = 4 ..... (ii)

Divide ii by i

$\frac{npq}{np} = \frac{4}{3}$

$q = \frac{4}{3}$, which is impossible because q lies between 0 to 1. It does not follow binomial distribution. (here mean is smaller than variance)

Additive (or reproductive) property of binomial distribution: If X and Y are two independent binomial variate. $X \sim B(n_1, p_1)$ and $Y \sim B(n_2, p_2)$, sum of two independent binomial variate does not follow binomial distribution, however if probability of success is same i.e $p_1 = p_2 = p$, then X + Y follows binomial distribution with parameter $n_1 + n_2$ and p i.e $X + Y \sim B(n_1 + n_2, p)$

Example : If X and Y are two independent binomial variate, $X \sim B(8, \frac{1}{4})$ and $Y \sim B(6, \frac{3}{4})$, what will be distribution of X+Y ?

Solution : Since probability of success is not same for X and Y. X + Y will not follow binomial distribution.

Example 5.6: If X and Y are two independent binomial variate, X~ B(8, $\frac{1}{3}$) and Y~ B(6, $\frac{1}{3}$), what will be distribution of X+Y? Find mean, mode, variance and P(X+Y = 2)

Solution: Since probability of success is same for both variate X and Y. X+Y will follow binomial distribution with parameters n= 8 + 6 = 14 and p = $\frac{1}{3}$

Mean = $14 \times \frac{1}{3} = \frac{14}{3} = 4.67$

Mode = $(14+1) \times \frac{1}{3} = 5$, since (n + 1)p is an integer, distribution is bimodal, two modes are $(5-1)$ and 5 i.e. 4 and 5 are two modes.

$p = \frac{1}{3}$, $q = \frac{2}{3}$

Variance = $14 \times \frac{1}{3} \times \frac{2}{3} = \frac{28}{9} = 3.11$

$P(X + Y = 2) = {}^{14}C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^{14-2} = 91 \times 0.000856 = 0.07793$

Recurrence relation for fitting the probabilities of binomial distribution: Suppose X follows binomial distribution with parameters n and p, probability mass function of random variable X is

$$P(X = x) = \binom{n}{x} p^x q^{n-x} \qquad (i)$$

$$P(X = x + 1) = \binom{n}{x+1} p^{x+1} q^{n-x-1} \qquad (ii)$$

Divide (ii) by (i)

$$\frac{P(X = x+1)}{P(X = x)} = \frac{\binom{n}{x+1} p^{x+1} q^{n-x-1}}{\binom{n}{x} p^x q^{n-x}} = \frac{\frac{n!}{(x+1)!(n-x-1)!}}{\frac{n!}{(x)!(n-x)!}} \times \frac{p^{x+1-x}}{q^{n-x-n+x+1}}$$

$$= \frac{n!}{(x+1)!(n-x-1)!} \times \frac{(x)!(n-x)!}{n!} \times \frac{p}{q} = \frac{n!}{(x+1)(x)!(n-x-1)!} \times \frac{(x)!(n-x)(n-x-1)!}{n!} \times \frac{p}{q}$$

$$= \frac{(n-x)}{(x+1)} \frac{p}{q}$$

where ${}^nc_{x+1} = \binom{n}{x+1} = \frac{n!}{(x+1)!(n-x-1)!}$ ${}^nc_x = \binom{n}{x} = \frac{n!}{(x)!(n-x)!}$

$P(X = x+1) = \frac{(n-x)}{(x+1)} \frac{p}{q} P(X = x)$ and $P(X=0) = q^n$

Here only P(X = 0) is to be computed and rest probabilities X = 1, 2, …, n is obtained by using recurrence relation.

Example: 150 sets of seeds were kept for germination. In each set there were 7 seeds. Number of seeds germinated in each set were recorded. Fit binomial distribution to the number of seeds germinated.

X	0	1	2	3	4	5	6	7
f	7	6	19	35	30	23	27	3

Solution:

X	**f**	**fx**	**(n-x)**	$\frac{n-x}{x+1}$	$\frac{n-x}{x+1}\frac{p}{q}$	**p**	**Expected frequency**
0	7	0	7	7.00	8.22	0.00436	0.65 ≅ 1
1	6	6	6	3.00	3.52	0.03580	5.37 ≅ 5
2	19	38	5	1.67	1.96	0.12610	18.92 ≅ 19
3	35	105	4	1.00	1.17	0.24716	37.01 ≅ 37
4	30	120	3	0.60	0.70	0.28918	43.45 ≅ 43
5	23	115	2	0.33	0.39	0.20242	30.61 ≅ 31
6	27	162	1	0.14	0.17	0.07894	11.98 ≅ 12
7	3	21				0.01342	2.01 ≅ 2
Total	150	567				1.00000	150

Mean $= \frac{567}{150} = 3.78$

$np = 3.78$

$p = \frac{3.78}{7} = 0.54$

$q = 1 - 0.54 = 0.46$

$\frac{p}{q} = \frac{0.54}{0.46} = 1.173913$

$P(X=0) = q^n = 0.46^7 = 0.004358$

$E(X) = N \times p$

Poisson distribution: Binomial distribution reduces to Poisson distribution when

i) $n \to \infty$

ii) $p \to 0$

iii) $np = \lambda$

A random variable X is having Poisson distribution with parameter λ and its probability mass function is given by

$$P(X = x) = \frac{\exp(-\lambda)\lambda^x}{x!} \qquad x = 0, 1, 2, 3, 4, \ldots$$

Example:

i) Number of accidents in some unit of time.

ii) Number of suicides in a city.

iii) Number of deaths due to heart attack in a city.

iv) Number of printing mistakes at each page of book.

v) The emission of radioactive particles

Example : A box contains 100 items and past experiences show that 2 % of the items in the box are defected. What is the probability that

i) box contains no defected items

ii) atleast one defected items

iii) atmost one defected items

iv) atleast two defected items

Solution: $p = \frac{2}{100} = 0.02$

$\lambda = np = 100 \times 0.02 = 2$

i) No defected items, $P(X = 0) = \frac{\exp(-2)2^0}{0!} = \exp(-2) = 0.1353$

ii) Atleast one defected items = $1 - P(X = 0)$, $1 - 0.1353 = 0.8647$

iii) Atmost one defected items = $P(X = 0) + P(X = 1)$

$P(X=1) = \frac{\exp(-2)2^1}{1!} = 2 \times 0.1353 = 0.2706$

Atmost one defected items = $0.1353 + 0.2706 = 0.4059$

iv) Atleast two defected items = $1 - P(X = 0) - P(X = 1)$

$= 1 - 0.1353 - 0.2706$

$= 0.5941$

Constants of Poisson distribution:

Binomial distribution reduced to Poisson distribution when

i) $n \to \infty$

ii) $p \to 0, q \to 1$

iii) $p = \lambda$

Hence mean and variance of Poisson distribution are obtained from Binomial distribution as

Constant	Binomial Distribution	Poisson Distribution
Mean	np	λ
μ_2	npq	$\lambda.1 = \lambda$

Note: In Poisson distribution mean and variance are same.

Mode: There are two cases

i) If λ is not an integer, then distribution is unimodal and integer part of λ will be value of mode

ii) If λ is an integer, then distribution is bi-modal two modes are $(\lambda-1)$ and λ

Example : Variance of Poisson distribution is 3, what will be mean and mode of the distribution

Solution: Mean = variance = $\lambda = 3$

Mode: Since $\lambda = 3$ is an integer, distribution is bi modal and two modes are $3-1 = 2$ and 3.

Example : Variance of Poisson distribution is 3.2. What will be mean and mode of the distribution

Solution: variance = $\lambda = 3.2$, In Poisson distribution mean and variance are same. Hence

Mean = 3.2

Mode: Since $\lambda = 3.2$ is not an integer, distribution is unimodal and mode is 3.

Example : A distribution is bi-modal at 2 and 3. Find mean and variance

Solution: Since distribution is bimodal and two modes are 2 and 3. Hence $\lambda = 3$

Example : If X and Y are two independent Poisson variate such that

P(X = 1) = P(X = 2) and P(Y = 2) = P(Y = 3), find E(X − 3Y) and V(X − 3Y)

Solution: Suppose λ is the parameter of X and μ is the parameter of Y

P(X =1) = P(X= 2)

$$\frac{\exp(-\lambda)\lambda^1}{1!} = \frac{\exp(-\lambda)\lambda^2}{2!}$$

$\lambda = 2$

$P(Y = 2) = P(Y = 3)$

$$\frac{\exp(-\mu)\mu^2}{2!} = \frac{\exp(-\mu)\mu^3}{3!}$$

$\mu = 3$

$E(X) = V(X) = 2$

$E(Y) = V(Y) = 3$

$E(X-3Y) = E(X) - 3E(Y) = 2 - 3 \times 3 = -7$

$V(X-3Y) = V(X) + (-3)^2 V(Y) = 2 + 9 \times 3 = 29$

Example: Probability of 3 successes = probability of 4 successes in Poisson distribution. Find mean, variance, mode.

Solution : $P(X = 3) = P(X = 4)$

Let λ is the parameter of Poisson distribution

$$\frac{\exp(-\lambda)\lambda^3}{3!} = \frac{\exp(-\lambda)\lambda^4}{4!}$$

$\lambda = 4$

Mean = variance = 4

Mode: Since $\lambda = 4$ is an integer, distribution is bi modal and two modes are 3 and 4

Example : $P(X = 2) = 9\ P(X = 4) + 90\ P(X = 6)$. Find parameter of Poisson distribution.

Solution:

Let λ is the parameter of Poisson distribution

$$\frac{\exp(-\lambda)\lambda^2}{2!} = 9\frac{\exp(-\lambda)\lambda^4}{4!} + 90\frac{\exp(-\lambda)\lambda^6}{6!}$$

$$\frac{\exp(-\lambda)\lambda^2}{2\times1} = 9\frac{\exp(-\lambda)\lambda^4}{4\times3\times2\times1} + 90\frac{\exp(-\lambda)\lambda^6}{6\times5\times4\times3\times2\times1}$$

$$\frac{\exp(-\lambda)\lambda^2}{2\times1} = 3\frac{\exp(-\lambda)\lambda^4}{8} + \frac{\exp(-\lambda)\lambda^6}{8}$$

$$\frac{\exp(-\lambda)\lambda^2}{2\times1} = \exp(-\lambda)\lambda^2\left(3\frac{\lambda^2}{8} + \frac{\lambda^4}{8}\right)$$

$4 = 3\lambda^2 + \lambda^4$

$\lambda^4 + 3\lambda^2 - 4 = 0$

$$\lambda^2 = \frac{-3\pm\sqrt{3^2-4(1)(-4)}}{2X1}$$

$$= \frac{-3\pm5}{2} = 1, -4$$

-4 is rejected as λ^2 is always positive

$\lambda = \pm 1$ or $\lambda = 1$ (-1 is rejected)

Example : A company manufacture insecticide and packed 500 bottles in a box and it is known that 0.1 % of the bottle in the box are defected. A company purchases 100 boxes, how many boxes will contain

i) no defected bottle

ii) atleast one defected bottle

Solution: Probability of defected bottle = $\frac{0.1}{100} = 0.001$

Mean defected bottle in a box $\lambda = 500 \times 0.001 = 0.5$

$P(X=0) = \exp(-0.5) = 0.6065$

N =100

Number of boxes containing no defected bottle = $100 \times 0.6065 \simeq 61$

Probability of atleast one defected bottle = $1 - P(X=0) = 1 - 0.6065 = 0.3935$

Number of boxes containing atleast one defected bottle = $100 \times 0.3935 \simeq 39$

Example : A book contains 200 pages and there are 100 mistakes in the book. A page is selected randomly, what is the probability that page contains

i) no mistake

ii) atleast one mistake

Solution:

Mean number of mistake in the page $\lambda = \frac{100}{200} = 0.5$

i) $P(X=0) = \exp(-0.5) = 0.6065$

ii) P(atleast one mistake) = $1 - P(X = 0) = 1 - 0.6065 = 0.3935$

Additive properties of Poisson distribution: If X and Y are two independent Poisson variate with parameter λ_1 and λ_2, then sum of two independent Poisson variate follows Poisson distribution with parameter $\lambda_1 + \lambda_2$

i.e. $(X + Y) \sim P(\lambda_1 + \lambda_2)$

Recurrence relation for fitting probabilities of Poisson distribution:

Suppose X is having Poisson distribution with parameter λ

$$P(X = x) = \frac{\exp(-\lambda)\lambda^x}{x!} \qquad \text{(i)}$$

$$P(X = x + 1) = \frac{\exp(-\lambda)\lambda^{x+1}}{x+1!} \qquad \text{(ii)}$$

Dividing (ii) by (i)

$$\frac{P(X = x + 1)}{P(X = x)} = \frac{\frac{\exp(-\lambda)\lambda^{x+1}}{(x+1)!}}{\frac{\exp(-\lambda)\lambda^{x}}{x!}} = \frac{\frac{\exp(-\lambda)\lambda^{x}\lambda}{(x+1)(x)!}}{\frac{\exp(-\lambda)\lambda^{x}}{x!}} = \frac{\lambda}{x+1}$$

$$P(X = x + 1) = \frac{\lambda}{x+1} P(X = x)$$

We have to find only $P(X = 0) = \exp(-\lambda)$ and rest probabilities is obtained by using recurrence relation.

Example : Fit Poisson distribution to the following set of data

X	0	1	2	3	4	5	6	7
f	123	62	33	24	3	4	1	0

Solution:

X	**f**	**fX**	$\frac{\lambda}{x+1}$	**p**	**Expected frequency**
0	123	0	0.95	0.38596	96.49 ≅ 96
1	62	62	0.48	0.36667	91.86 ≅ 92
2	33	66	0.32	0.17600	43.73 ≅ 44
3	24	72	0.24	0.05632	13.88 ≅ 14
4	3	12	0.19	0.01352	3.30 ≅ 3
5	4	20	0.16	0.00257	0.63 ≅ 1
6	1	6	0.14	0.00041	0.10 ≅ 0
7	0	0		0.00006	0.01 ≅ 0
Total	250	238		1.00	250.00

Mean $\lambda = \frac{238}{250} = 0.952$

$P(X=0) = \exp(-0.952) = 0.386$

Normal Distribution:

Binomial distribution and Poisson distributions reduces to normal distribution:

In case of binomial distribution,

Binomial distribution reduces to normal distribution when

i) Number of trial is very large i.e. $n \to \infty$

ii) $p \approx q$

In case of Poisson distribution

Poisson distribution reduces to normal distribution when $\lambda \to \infty$

A random variable X is having normal distribution with parameter mean = μ and variance = σ^2 and its probability density function is given by

$$f(X: \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left\{-\frac{1}{2}\left(\frac{X-\mu}{\sigma}\right)^2\right\} \qquad -\infty < X < +\infty\,,\ -\infty < \mu < +\infty\,,\ \sigma > 0$$

Standard normal variate: To find the probability between two points in a normal distribution, random variable X is converted into standard normal variate Z, where

$$Z = \frac{X-\mu}{\sigma}$$

If $X \sim N(\mu, \sigma^2)$, then $Z \sim N(0, 1)$

Proof: $E(Z) = E\left(\frac{X-\mu}{\sigma}\right) = \frac{1}{\sigma}E(X - \mu)$

$$= \frac{1}{\sigma}\ (E(X) - E(\mu))$$

$$= \frac{1}{\sigma}\ (\mu - \mu) = 0$$

$$V(Z) = V\left(\frac{X-\mu}{\sigma}\right)$$

$$= \frac{1}{\sigma^2} V(X - \mu)$$

$$= \frac{1}{\sigma^2} [V(X) + V(\mu)]$$

$$= \frac{1}{\sigma^2}(\sigma^2+0) = 1$$

Standard normal table is available to obtain the probability between 0 to z. When $X = \mu$, value of $Z = 0$, when X is less than mean, Z is negative and when X is greater than mean , Z is positive. The curve is symmetrical at mean, hence 0.5 probability is towards the right hand side and 0.5 probability is towards the left hand side of the curve.

Note: To find probabilities between two points

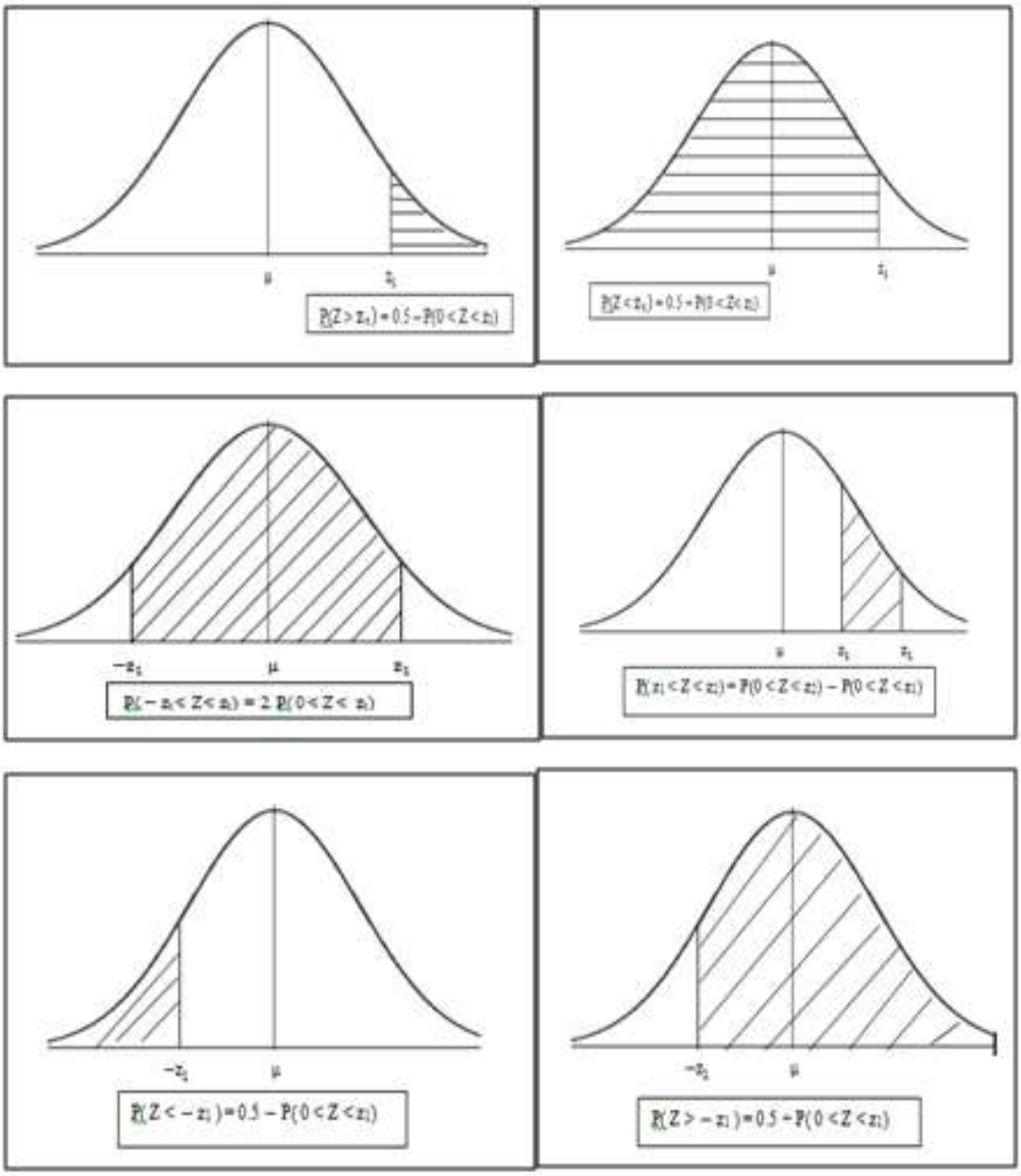
i) $P(Z > z_1) = 0.5 - P(0 < Z < z_1)$

ii) $P(Z < z_1) = 0.5 + P(0 < Z < z_1)$

iii) $P(-z_1 < Z < z_1) = 2.\ P(0 < Z < z_1)$

iv) $P(z_1 < Z < z_2) = P(0 < Z < z_2) - P(0 < Z < z_1)$

v) $P(Z < -z_1) = 0.5 - P(0 < Z < z_1) P(Z > -z_1) = 0.5 + P(0 < Z < z_1)$



Chief characteristics of normal distribution:

i) The curve is bell shaped and symmetrical at mean

ii) Mean, median and mode are same.

iii) All odd order moments are 0

iv) As X increases f(x) decreases rapidly

v) Since f(x) represents probability, no portion of the curve lies below X axis

vi) Quartile deviation, mean deviation and standard deviation are in the ratio of 10:12:15

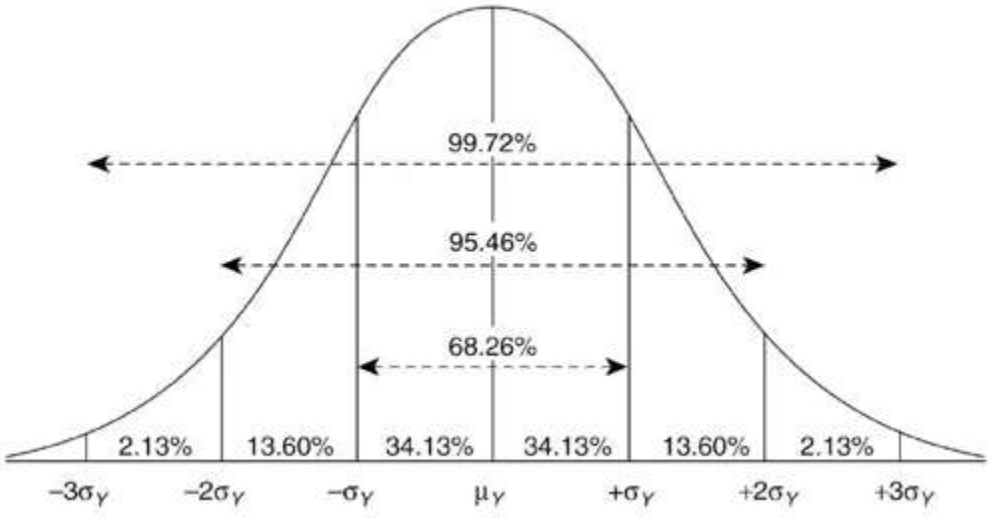
vii) Linear combination of independent normal variate is also normally distributed

viii) Area properties

a) $P(\mu - \sigma < X < \mu + \sigma) = 0.6826$

b) $P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9544$

c) $P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$



Example : If X is normally distributed with mean = 50 and variance = 16 find the probability that

i) $P(X > 54)$

ii) $P(X < 54)$

iii) $P(46 < X < 54)$

iv) $P(X < 46)$

v) $P(X > 46)$

vi) $P(54 < X < 58)$

Solution: $X \sim N(\mu = 50, \sigma^2 = 16), \sigma = 4$

i) when $X = 54$

$$Z = \frac{54 - 50}{4} = 1$$

$P(X > 54) = P(Z > 1)$

$= 0.5 - P(0 < Z < 1)$

$= 0.5 - 0.3413$

$= 0.1587$

ii) $P(X < 54) = P(Z < 1)$

$= 0.5 + P(0 < Z < 1)$

$= 0.5 + 0.3413 = 0.8413$

iii) $P(46 < X < 54)$

When $X = 46$

$Z = \frac{46 - 50}{4} = -1$

$P(46 < X < 54) = P(-1 < Z < 1)$

$= P(-1 < Z < 0) + P(0 < Z < 1)$

$= 2.\ P(0 < Z < 1)$ (Since the curve is symmetrical)

$= 2 \times 0.3413 = 0.6826$

iv) $P(X < 46) = P(Z < -1) = 0.5 - P(-1 < Z < 0)$

$= 0.5 - 0.3413 = 0.1587$

v) $P(X > 46) = P(Z > -1) = 0.5 + P(-1 < Z < 0)$

$= 0.5 + 0.3413 = 0.8413$

vi) $P(54 < X < 58)$

When $X = 58$

$Z = \frac{58 - 50}{4} = 2$

$P(54 < X < 58) = P(1 < Z < 2)$

$= P(0 < Z < 2) - P(0 < Z < 1)$

$= 0.4772 - 0.3413 = 0.1259$

Example : Weight of fruit is normally distributed with mean weight 60 g and variance 9. Fruits whose weight lies between 57 to 66 g are sold @ Rs 8 per fruit, fruit whose weight is < 57 g are sold @ Rs 6 per fruit and rest are sold @ Rs 10 per fruit. Find total amount collected if in all there are 10000 fruits.

Solution: $X \sim N(\mu = 60, \sigma^2 = 9)$, $\sigma = 3$

When $X = 57$

$Z = \frac{57 - 60}{3} = -1$

When $X = 66$

$Z = \frac{66 - 60}{3} = 2$

$P(57 < X < 66) = P(-1 < Z < 2)$

$= P(-1 < Z < 0) + P(0 < Z < 2)$

$= 0.3413 + 0.4772 = 0.8185$

$P(X < 57) = P(Z < -1) = 0.5 - P(0 < Z < 1) = 0.5 - 0.3413 = 0.1587$

Probability of rest fruits = 1 – 0.8185 – 0.1587 = 0.0228

Total fruits = 10000

Number of fruits whose weight lies between 57 to 66 = 10000 × 0.8185 = 8185

Number of fruits whose weight is less than 57 =10000 × 0.1587 = 1587

Rest fruits = 10000 × 0.0228 = 228

Weight of fruit (g)	Number of fruits	Price (Rs)	Amount (Rs)
57 to 66	8185	8	65480
< 57	1587	6	9522
Rest fruits	228	10	2280
Total	10000		77282

Total amount collected = 77282

Linear combination of independent normal variate: If X and Y are two independent normal variate. $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_{2,}^2)$, then

$E(aX + bY) = aE(X) + bE(Y)$

$V(aX + bY) = a^2V(X) + b^2V(Y)$

If $a = 1, b = 1$

$E(X + Y) = E(X) + E(Y) = \mu_1 + \mu_2$

$V(X + Y) = V(X) + V(Y) = \sigma_1^2 + \sigma_2^2$

If $a = 1, b = -1$

$E(X - Y) = E(X) - E(Y) = \mu_1 - \mu_2$

$V(X - Y) = V(X) + V(Y) = \sigma_1^2 + \sigma_2^2$

Additive properties of normal distribution: If X and Y are two independent normal variate and $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_{2,}^2)$, then sum as well as difference of two independent normal variate are also normally distributed i.e.

$(X + Y) \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

$(X - Y) \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$

Example : If X and Y are two independent normal variate, X ~N(20, 9) and Y~N(10,4), what will be the distribution of i) 2X + 3Y ii) 2X – 3Y iii) X – Y iv) X + Y

i) 2X + 3Y

$E(2X + 3Y) = 2E(X) + 3E(Y) = 2 \times 20 + 3 \times 10 = 70$

$V(2X + 3Y) = 4V(X) + 9V(Y) = 4 \times 9 + 9 \times 4 = 72$

$2X + 3Y \sim N(70, 72)$

ii) 2X – 3Y

$E(2X - 3Y) = 2E(X) - 3E(Y) = 2 \times 20 - 3 \times 10 = 10$

$V(2X - 3Y) = 4V(X) + 9V(Y) = 4 \times 9 + 9 \times 4 = 72$

$2X - 3Y \sim N(10, 72)$

iii) $X - Y$

$E(X - Y) = E(X) - E(Y) = 20 - 10 = 10$

$V(X - Y) = V(X) + V(Y) = 9 + 4 = 13$

$X - Y \sim N(10, 13)$

iv) X + Y

$E(X + Y) = E(X) + E(Y) = 20 + 10 = 30$

$V(X - Y) = V(X) + V(Y) = 9 + 4 = 13$

$X + Y \sim N(30, 13)$

Example : Fruit yield of plant is normally distributed with mean number of fruits 600 and variance 1600. If there are 1000 plants in an orchard, how many plants have i) mean fruit yield less than 700 ii) between 650 to 700 iii) more than 650

Solution: $X \sim N(600, 1600)$ $\sigma = 40$

When X = 700

$Z = \frac{700 - 600}{40} = 2.5$

i) $P(X < 700) = P(Z < 2.50)$

$= 0.5 + P(0 < Z < 2.5) = 0.5 + 0.4938 = 0.9938$

Number of plants having fruit yield less than 700 = 1000 × 0.9938 = 994

ii) $P(650 < X < 700)$

When X = 650

$Z = \frac{650 - 600}{40} = 1.25$

$P(650 < X < 700) = P(1.25 < Z < 2.50) = P(0 < Z < 2.50) - P(0 < Z < 1.25)$

$= 0.4938 - 0.3944 = 0.0994$

Number of plants having fruit yield between 650 to 700 = $1000 \times 0.0994 = 99$

iii) $P(X > 650) = P(Z > 1.25) = 0.5 - P(0 < Z < 1.25) = 0.5 - 0.3944 = 0.1056$

Number of plants having fruit yield more than $650 = 1000 \times 0.1056 = 106$

Example : 1000 students are studying the course of statistics in the university. Probability that a student require Statistics book from the library is 0.45. How many books should be available in the library such that probability of getting issue of a book from library is atleast 0.8

Solution: Here n = 1000, p = 0.45 q = 0.55, n is large and $p \approx q$, binomial distribution reduces to normal distribution

Mean = np = $1000 \times 0.45 = 15.73$

Variance = npq = $1000 \times 0.45 \times 0.55 = 247.5$

Standard Deviation $\sigma = 15.73$

From the standard normal table area from $-\infty$ to + 0.84 is 0.80 i.e from 0.84 to $+\infty$ is 0.20

Let X represents number of books available in the library

$\frac{X - \mu}{\sigma} > 0.84$

$\frac{X - 450}{15.73} > 0.84$

$X - 450 > 0.84 \times 450$

$X > 13.22 + 450$

$X > 463.22$

So library must have atleast 464 books in the library

Fitting of Normal distribution: Suppose X is normally distributed with mean = μ and variance = σ^2 . We assume that first class is less than upper limit of that class and last class as more than lower limit of that class, then convert all upper limit into standard normal variate Z.

Example : Height (feet) of 1000 trees was measured and following is the distribution of height of trees.

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100	Total
f	2	8	38	96	145	272	220	172	44	3	1000

Solution:

Class (Height)	**f**	**MV (X)**	**d = $\frac{X-45}{10}$**	**fd**	**fd^2**	**Z**	**Probability from 0 to z**	**p**	**Expected Frequency**
0-10	2	5	-4	-8	32	-3.07	0.4989	0.0011	1
10-20	8	15	-3	-24	72	-2.42	0.4922	0.0067	7
20-30	38	25	-2	-76	152	-1.77	0.4616	0.0306	31
30-40	96	35	-1	-96	96	-1.12	0.3686	0.0930	93
40-50	145	45	0	0	0	-0.47	0.1808	0.1878	188
50-60	272	55	1	272	272	0.19	0.0754	0.2562	256
60-70	220	65	2	440	880	0.84	0.2996	0.2242	224
70-80	172	75	3	516	1548	1.49	0.4319	0.1323	132
80-90	44	85	4	176	704	2.14	0.4838	0.0519	52
90-100	3	95	5	15	75			0.0162	16
Total	1000			1215	3831			1	1000

Mean $= 45 + 10 \times \frac{1215}{1000} = 45 + 12.15 = 57.15$

Standard Deviation = $\sqrt{\frac{10^2}{1000}[3831 - \frac{1215^2}{1000}]} = \sqrt{\frac{10^2}{1000}[3831 - 1476.225]}$

$= \sqrt{\frac{10^2}{1000} \times 2354.775} = \sqrt{235.48} = 15.35$

$Z = \frac{10 - 57.15}{15.35} = -3.07$

$P(0 < X < 10) = P(X < 10) = P(Z < -3.07) = 0.5 - 0.4989 = 0.0011$

$Z = \frac{20 - 57.15}{15.35} = -2.42$

$P(10 < X < 20) = P(-3.07 < Z < -2.42) = 0.4989 - 0.4922 = 0.0067$

$Z = \frac{30 - 57.15}{15.35} = -1.77$

$P(20 < X < 30) = P(-2.42 < Z < -1.77) = 0.4922 - 0.4616 = 0.0306$

$Z = \frac{40 - 57.15}{15.35} = -1.12$

$P(30 < X < 40) = P(-1.77 < Z < -1.12) = 0.4616 - 0.3686 = 0.0930$

$Z = \frac{50 - 57.15}{15.35} = -0.47$

$P(40 < X < 50) = P(-1.12 < Z < -0.47) = 0.3686 - 0.1808 = 0.1878$

$Z = \frac{60 - 57.15}{15.35} = 0.19$

$P(50 < X < 60) = P(-0.47 < Z < 0.19) = P(-0.47 < Z < 0) + P(0 < Z < 0.19)$

$= 0.1808 + 0.0754 = 0.2562$

$Z = \frac{70 - 57.15}{15.35} = 0.84$

$P(60 < X < 70) = P(0.19 < Z < 0.84) = 0.2996 - 0.0754 = 0.2242$

$Z = \frac{80 - 57.15}{15.35} = 1.49$

$P(70 < X < 80) = P(0.84 < Z < 1.49) = 0.4319 - 0.2996 = 0.1323$

$Z = \frac{90 - 57.15}{15.35} = 2.14$

$P(80 < X < 90) = P(1.49 < Z < 2.14) = 0.4838 - 0.4319 - = 0.0519$

$P(90 < X < 1000) = P(X > 90) = P(Z > 2.14) = 0.5 - 0.4838 = 0.0162$

Importance of Normal distribution:

i) Most of the distribution occurring in practice i.e. binomial, Poisson and other distributions can be approximated by normal distribution

ii) Most of the sampling distributions i.e. t, F and $\chi 2$ approaches to normal distribution for large sample

iii) The distribution of sample statistic i.e. sample mean, variance approaches to normal distribution for large sample.

iv) Entire theory of small sample test i.e. t, F and $\chi 2$ is based on the fundamental assumption that parent population from which the sample have been drawn is normally distributed.

v) Normal distribution has wide applications in quality control

vi) If variable X is not normally distributed, then by suitable transformation, the variable reduces to normal distribution. If variable X is skewed distribution, then square root of X is normally distributed.

In case of percentage data, where percentage is calculated from the count, data can be easily transformed to normal data by using suitable transformations.

a) If percentage lies between 30 -70, data is normally distributed, hence no transformation is used

b) If percentage values lie between 0 to 30 or 70 to 100, square root transformation is used.

c) If both conditions are not fulfilled, angular transformation is used.

Gamma Distribution : A continuous random variable X is having Gamma distribution with parameter λ and its probability density function is given by

$f(x) = \frac{exp^{-x}\, x^{\lambda-1}}{\Gamma\lambda}$; $x > 0$, $\lambda > 0$

Constant of Gamma distribution

Mean = λ

$\mu_2 = \lambda$

Note :

- Mean and variance of Gamma distribution are same